AG-001

Designing Heated Laminated Glass

Design guidance for engineers and OEMs specifying embedded-conductor heated transparencies.

1. Start With the Requirement, Not the Conductor

Every heated transparency program resolves to four inputs: the clearing or anti-fog requirement, the available supply voltage, the aperture geometry, and the optical acceptance criteria. Conductor format is an outcome of those inputs, not a starting assumption.

2. Voltage Comes First

Design input	Effect on the conductor design
Supply voltage	Sets target circuit resistance for a given power density ($P = V^2 / R$).
Power density	Drives clearing time and constrains interlayer and seal temperature.
Run length	Determines resistance per conductor path and bus bar placement.
Pitch	Trades thermal uniformity against optical signature and total resistance.
Wire diameter	Trades resistance and current capacity against perceived visibility.

3. Conductor Format Selection

- Embedded wire — high power density, curved-part friendly, low sheet resistance, discrete visual signature.
- Conductive mesh — area conduction with shielding capability; transmission trade-off must be accepted.
- ITO film and ITO glass — uniform, visually clean; power density and sheet resistance differ from embedded wire and are evaluated per program.
- Hybrid builds — film for uniform area conduction with wire for low-resistance distribution or local high-power zones.

4. Bus Bars and Terminations

A common failure point in heated glass is the termination rather than the heated field. Size the bus bar cross-section for the full circuit current, keep conductor runs perpendicular to the bus bars so path lengths are comparable, and control joint quality, concealment, and lead exit sealing as design features rather than assembly details.

5. Zoning and Exclusion Areas

- Zone camera apertures, wiper park, and primary vision separately to reduce total power draw.
- Define exclusion areas for ADAS fields of view, antennas, and sensor windows early.
- Evaluate interaction between heating grids, shielding layers, and any embedded antenna elements.

6. Sensing and Overheat Protection

Where the build can exceed interlayer or seal limits, place an embedded sensing element at the point expected to reach the highest temperature within the controlled zone rather than at a convenient edge. HLG embeds the sensing element within the interlayer rather than placing it on the surface, in order to provide a more stress-free laminate.

7. Validation Path

- Coupon-level circuit and lamination verification.
- Representative-size samples for uniformity and clearing time.
- Energized optical evaluation per HLG-QS-1724.
- Power cycling and environmental exposure on the frozen build.
- Process and lot-traceability freeze before production release.

Clearing time is established by test on representative samples under defined ambient conditions — it is not predicted from the circuit design alone.

This document is provided for engineering information. Descriptions reflect development and evaluation capabilities and are not statements of certification, approval, or guaranteed performance. Application-specific performance is established by test on representative samples.